

Brandon Reese

Senior Data | Platform Engineer

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SUMMARY

Senior Data Engineer with 10+ years driving **enterprise-scale, modern data platforms** and **analytics-ready pipelines** across finance, enterprise, and precision medicine. Expert in **Python, SQL, Snowflake, Spark, dbt, Airflow, and Kafka**, delivering **multi-terabyte ingestion, transformation, and ML feature pipelines** that cut latency, improve reliability, and reduce compute costs.

Proven in **data validation, regression testing, and anomaly detection**, making **architecture and workflow decisions** that ensure **compliant, research-ready, high-quality datasets**. Trusted collaborator with ML, research, and business teams, and mentor to engineers, enabling **actionable, data-driven insights** across complex, multi-domain environments.

SKILLS

Programming & Scripting: Python, SQL, Scala, Shell Script

Databases & Data Warehousing: Snowflake, Redshift, BigQuery, PostgreSQL, MongoDB, Oracle, Delta Lake, Iceberg

Data Transformation & Modeling: dbt, Dimensional Modeling, ELT/ETL Development, Analytics-Ready Datasets, Modular Transformations

Pipeline Orchestration & Streaming: Apache Airflow, Dagster, Kafka, Fivetran, Event-Driven Ingestion, Batch & Streaming Pipelines

Big Data & Distributed Processing: Apache Spark / PySpark, High-Volume Data Processing, Performance Optimization

Cloud & Infrastructure: AWS (S3, Glue, Lambda, EMR, CloudFormation, IAM, CloudWatch), Terraform, Kubernetes, CI/CD (GitHub Actions)

Data Governance, Observability & Testing: Unity Catalog, Data Validation, Regression Testing, Anomaly Detection, Prometheus, Grafana, SLA Monitoring

BI & Analytics Enablement: Looker, Tableau, Power BI, Sigma, Metrics Standardization, Self-Service Dashboards

Machine Learning & AI Pipelines: ML Feature Engineering, Scikit-learn, TensorFlow, NumPy, Pandas, Model-Ready Dataset, MLOps

EXPERIENCE

Tempus – Chicago, IL

Dec 2021 – Present

Senior Data Engineer

- Architected **Snowflake-based data warehouses** and modular **dbt transformation pipelines** to harmonize multi-modal clinical, genomic, and imaging data from 100+ healthcare institutions, embedding **automated schema tests and regression validation**, reducing downstream data inconsistencies by ~40% and enabling reliable research-ready datasets.
- Developed **real-time ingestion frameworks** using **Kafka** and orchestrated batch & streaming workflows with **Airflow**, enabling sub-30-minute updates of lab results, patient encounters, and clinical documents while making critical decisions on **partitioning, SLA adherence, and error handling** for research-critical datasets.

- Built **Spark/PySpark pipelines** processing multi-terabyte daily datasets, implementing **row-level validation, anomaly detection, and incremental load strategies**, which decreased downstream ML feature errors by ~35% and improved pipeline reliability across oncology and cardiology research projects.
- Designed **analytics-ready feature tables** for ML and research teams, transforming raw clinical and molecular datasets into model-ready formats, enabling predictive oncology models to retrain automatically with reduced data preparation time (~50% faster).
- Implemented **data validation frameworks** in Python and SQL, including automated schema enforcement, null/duplicate checks, and anomaly alerts, preventing invalid records from entering AI pipelines and research dashboards, improving **data trustworthiness** by ~40%.
- Led **architecture decisions** for cross-institutional data pipelines, balancing **data freshness, throughput, and cost**, and collaborated with clinical and ML teams to validate feature correctness before production deployment.
- Built **observability dashboards** using Prometheus and Grafana for ingestion, transformation, and ML feature pipelines, enabling SLA monitoring, anomaly detection, and automated alerts, reducing pipeline incident response time by ~60%.
- Optimized **Snowflake query performance** through clustering, micro-partitioning, and caching strategies, cutting query runtime by ~25% and compute costs by ~20% while maintaining high throughput for researcher queries across petabyte-scale datasets.
- Coordinated **incremental loads and historical backfills** for long-term patient datasets, ensuring idempotency and zero downtime while maintaining compliance with HIPAA and institutional policies.
- Partnered with **data science and clinical research teams** to define KPIs, data contracts, and feature pipelines for predictive models, making critical decisions on transformation logic, aggregation strategies, and validation approaches.
- Implemented **CI/CD pipelines** using GitHub Actions, Terraform, and Docker to containerize workflows for reproducible deployment across cloud environments, reducing deployment failures by ~50% and maintaining regulatory compliance.
- Developed **self-service research datasets and cohort-building frameworks**, enabling clinical and research teams to query patient populations, test hypotheses, and generate predictive insights independently, decreasing ad-hoc requests by ~45%.
- Participated in **multi-domain platform modernization**, integrating genomic, imaging, and clinical datasets into unified schemas to power precision medicine insights across oncology, cardiology, and neurology.
- Continuously tuned **ETL and transformation workflows**, introducing parallelization, caching, and workload isolation strategies to handle scaling demands while maintaining strict compliance and data privacy standards.
- Mentored junior engineers on **Python scripting, SQL optimization, dbt best practices, and pipeline architecture**, fostering team capability to manage high-volume, multi-domain data projects and accelerating onboarding.

Wipro – Dallas, TX

Sep 2018 – Nov 2021

Data Engineer

- Designed and implemented **Snowflake-based data warehouses** for multiple enterprise clients, orchestrating ETL pipelines with **Airflow** and modular **Python scripts**, reducing batch processing latency by ~40% and enabling near-real-time analytics for Finance, Retail, and Healthcare teams.
- Migrated legacy Redshift pipelines to **dbt transformation models**, introducing automated **unit testing, version control**, and modular SQL logic, improving **data quality** by ~35% and standardizing datasets across multi-domain clients.

- Built **Spark/PySpark workflows** on EMR for processing high-volume datasets (~1–2TB/day), tuning memory allocation, partitioning, and transformation logic to reduce job failures by ~45% while meeting strict SLA windows.
- Developed **event-driven ingestion frameworks** using **Kafka**, handling near-real-time operational data for marketing and transactional analytics, reducing reporting latency from hours to ~15 minutes.
- Designed **dimensional modeling** in Snowflake and prepared **analytics-ready datasets**, enabling BI teams to self-serve KPIs, decreasing ad-hoc data requests by ~40% and aligning dashboards with verified **business rules**.
- Implemented **data validation and anomaly detection frameworks** using Python and SQL, embedding schema enforcement, row-level checks, and alerting pipelines to prevent corrupt data from reaching dashboards, reducing production errors by ~30%.
- Partnered with product and analytics teams to define **data contracts and KPIs**, making architecture and transformation decisions, validating outputs through **staged QA pipelines** and regression tests for multi-client consistency.
- Introduced **CI/CD pipelines** with GitHub Actions, Terraform, and Docker, containerizing workflows for reproducible deployments across client environments, reducing deployment failures by ~50% and standardizing multi-team development.
- Optimized **Snowflake warehouse performance** with clustering, micro-partitioning, and workload isolation, decreasing compute costs by ~25% while maintaining high throughput for peak reporting and analytics periods.
- Built **observability dashboards** using Prometheus and Grafana, defining SLA thresholds, alerting, and anomaly tracking, improving on-call incident response time by ~60% and providing actionable monitoring for client stakeholders.
- Conducted **root-cause analysis** on pipeline failures, tuning Spark/PySpark jobs, Python transformation logic, and SQL queries to improve reliability across batch and streaming jobs.
- Mentored junior engineers on **modular pipeline design, SQL optimization, Python scripting**, and best practices for transformation logic, increasing team efficiency and accelerating onboarding for multiple enterprise clients.
- Coordinated incremental load strategies and **historical backfills** for client datasets, ensuring idempotency, data integrity, and zero downtime, even for datasets spanning multiple years.
- Contributed to **ML feature pipelines** by collaborating with data science teams, preparing Snowflake datasets, validating feature correctness, and supporting automated retraining for predictive models.
- Led cross-team **performance tuning initiatives**, benchmarking ETL runtime, refining Spark and SQL job parameters, and implementing parallelization strategies, reducing total pipeline runtime by ~30% on average across client projects.
- Worked closely with client stakeholders to **define reporting requirements, validate KPIs, and ensure dashboards reflected actionable insights**, balancing engineering trade-offs between data freshness, query performance, and compute costs.

Robinhood – Menlo Park, CA

Jun 2016 – Jul 2018

ETL Engineer

- Built and maintained **ETL pipelines** for brokerage and transactional data, applying Python scripts and **SQL transformations** to reconcile trade records, reducing reconciliation lag from 4 hours to under 40 minutes during volatile trading sessions.
- Refactored legacy batch jobs into **Spark workflows on EMR**, tuning **Python scripts** and **SQL queries** to process 500–700GB of daily trade and market data, reducing runtime by ~30% and stabilizing jobs during peak trading periods.
- Designed **data validation checks** with row-level anomaly detection, schema enforcement, and null thresholds, cutting downstream reporting errors by ~35% and improving audit reliability.

- Partnered with Risk and Compliance to implement a **trade reconciliation framework** using SQL-based audits and deterministic matching logic, detecting low-frequency mismatches before financial close and enabling faster regulatory reporting.
 - Modeled reporting datasets in **Redshift** with dimensional schemas, improving BI dashboard performance by ~25% and reducing recurring data requests from business teams.
 - Introduced **lightweight CI validation tests** for ETL workflows, combining Python unit tests and SQL transformation checks, decreasing post-deployment hotfixes during high-traffic trading days.
 - Collaborated with Product Analytics to define **event tracking contracts**, implementing validation scripts to ensure consistent metrics across dashboards during rapid feature rollouts.
 - Developed **incremental load patterns** using staging tables, enabling safe backfills of up to 18 months of historical trade data without overwriting reconciled records.
 - Conducted root-cause analysis of **Spark executor failures**, tuning Python jobs and SQL queries after benchmarking peak trade datasets, reducing failure rates by ~50%.
 - Built monitoring dashboards with **CloudWatch metrics** and query performance logging, setting thresholds based on historical baselines to reduce false-positive alerts by ~40%.
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EDUCATION

Bachelor of Science in Computer Science
University of North Florida, Jacksonville, Florida
Aug 2012 – May 2016